mentioned earlier, one that is shaped like a cube but goes on forever, with poppy seeds playing the role of galaxies. As the muffin bakes, the dough expands, causing each poppy seed to rush away from the others.) This shape is called *flat space* because, unlike the spherical example, it has no curvature (a meaning of "flat" that mathematicians and physicists use, but that differs from the colloquial meaning of "pancake-shaped.")[11]

One nice thing about both the spherical and the infinite flat shapes is that you can walk endlessly and never reach an edge or a boundary. This is appealing because it allows us to avoid thorny questions: What is beyond the edge of space? What happens if you walk into a boundary of space? If space has no edges or boundaries, the question has no meaning. But notice that the two shapes realize this attractive feature in different ways. If you walk straight ahead in a spherically shaped space, you'll find, like Magellan, that sooner or later you return to your starting point, never having encountered an edge. By contrast, if you walk straight ahead in infinite flat space, you'll find that, like the Energizer Bunny, you can keep going and going, again never encountering an edge, but also never returning to where your journey began. While this might seem like a fundamental difference between the geometry of a curved and a flat shape, there is a simple variation on flat space that strikingly resembles the sphere in this regard.

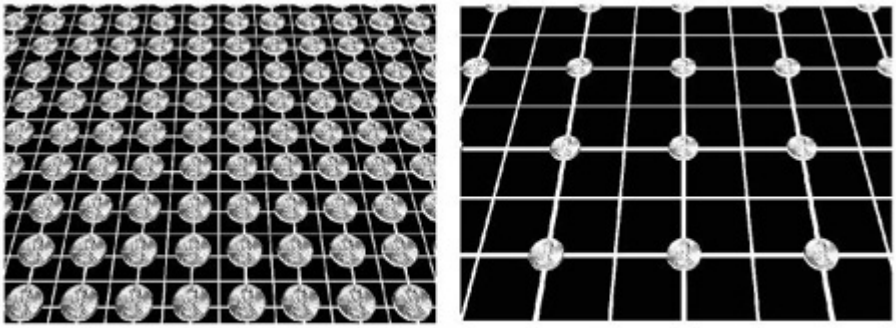


Figure 8.4 **(**a**)** The view from any penny on an infinite flat plane is the same as the view from any other. **(**b**)** The farther apart two pennies are in Figure 8.4a, the greater the increase in their separation when the plane expands.